ASSESSMENT

Program – CompTia N+

Code – M2-A1

Total Time – 3 Hours

Stack – Networking Hardware

SECTION A – CONCEPT APPLICATION

SCENARIO:- Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Question:- What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

- To ensure good performance and hardware compatibility, Nexus Innovations should consider the following factors while selecting the CPU, RAM, and storage for employees who use programming software, web browsers, and video conferencing applications simultaneously.

- 1. CPU (Central Processing Unit) :- Select a modern multi-core processor, such as Intel Core i5/i7 or AMD Ryzen 5/7, for multitasking.
- Check the processor's clock speed and core/thread count for smooth performance.
- Ensure the CPU socket is compatible with the motherboard.
- Check motherboard chipset support, BIOS compatibility, and CPU power requirements.

- 2. RAM (Random Access Memory) :- Choose at least 16 GB RAM for comfortable multitasking; 32 GB is better for heavy programming or running virtual machines.
- Prefer dual-channel RAM, such as 2 × 8 GB, if supported by the motherboard.

- Check RAM type (DDR4 or DDR5), supported speed, capacity, and voltage.
- Ensure the RAM is compatible with the motherboard and CPU memory controller.

- 3. Storage :- Prefer an NVMe SSD for fast booting, application loading, and file transfers.
- Select at least 512 GB storage; 1 TB is better if employees store large projects and files.
- Check whether the motherboard supports the selected NVMe or SATA drive and the required M.2 slot or connector.
- Consider additional storage or backup solutions for important company data.

SCENARIO :- After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question :- Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

- After installing antivirus software, the office computers may become slower because the antivirus uses system resources to scan files, monitor programs, and detect malware.

- 1. CPU (Central Processing Unit) :- Antivirus software uses CPU processing power to scan files and check running applications.
- Real-time protection continuously monitors system activities.
- During full system scans, CPU usage may increase, making other applications slower.

- 2. RAM (Random Access Memory) :- Antivirus software uses RAM to run background services, store scanning data, and maintain security processes.
- If RAM is limited, less memory remains available for other applications.
- This can cause slow multitasking, application freezing, or delays when switching between programs.

- 3. Storage (HDD/SSD) :- Antivirus scans read many files from the storage drive, increasing disk activity.
- On an HDD, frequent reading and seeking can make the system feel slower.
- If storage usage is very high or the drive is nearly full, application loading and file operations may become slower.

- Solutions :-
- Schedule full antivirus scans outside office hours.
- Keep antivirus software updated and configure real-time protection properly.
- Close unnecessary background applications.

- Upgrade RAM if memory usage is consistently high.
- Use an SSD instead of an HDD if storage performance is a bottleneck.

SCENARIO :- A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question :- Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

- If a USB keyboard and printer work properly on one computer but not on another, the problem may be caused by incompatible drivers, outdated operating system files, or incorrect USB settings. I would follow these steps:
- 1. Check the USB Connection :-
- Disconnect and reconnect the keyboard and printer.
- Try different USB ports on the affected computer.
- Restart the computer and check whether the devices work.

- 2. Check Device Manager :-
- Open Device Manager by right-clicking the Start button.
- Look for the keyboard and printer under their respective categories.
- Check for a yellow warning symbol, which may indicate a driver problem.

- 3. Update or Reinstall Drivers :-
- Right-click the affected device and select Update driver.
- Download the correct driver from the manufacturer's official website if needed.
- If the driver is corrupted, uninstall the device and restart the computer so Windows can reinstall it.

- 4. Check Operating System Compatibility :-
- Confirm that the device supports the installed Windows version.
- Install pending Windows updates and restart the computer.
- Check whether required system services, such as the Print Spooler service for printers, are running.

- 5. Test the Devices Again :-
- Reconnect the keyboard and printer.
- Test keyboard input and print a test page.
- If the problem continues, compare the affected computer's drivers and Windows settings with those of the working computer.

SCENARIO :- Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question :- Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

- Nexus Innovations experiences frequent power cuts and voltage fluctuations. A UPS (Uninterruptible Power Supply) and a surge protector help protect computer components and reduce the risk of data loss.

- 1. UPS (Uninterruptible Power Supply) :-
- A UPS provides temporary battery power when the main electricity supply fails.
- It gives employees time to save their work and shut down computers safely.
- It helps prevent data loss and file corruption caused by sudden power cuts.
- Some UPS models also regulate voltage to protect computers from certain voltage fluctuations.
- It keeps essential equipment running temporarily, depending on battery capacity and power load.

- 2. Surge Protector :-
- A surge protector protects computer hardware against sudden increases in voltage, called power surges.
- It diverts excess voltage away from connected equipment, depending on its protection design.
- It helps protect sensitive components such as the motherboard, CPU, RAM, and storage devices.
- It reduces the risk of hardware damage caused by lightning-related or electrical switching surges.

- 3. Recommended Solution :-
- Nexus Innovations should use a suitable UPS with adequate battery capacity and surge protection. Employees should save their work regularly and shut down systems safely during extended power cuts. Surge protectors should be properly rated and grounded.

SCENARIO :- New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question :- Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

- At Nexus Innovations, employees can access internal systems using IP addresses but cannot access them by name. Sometimes, their devices also fail to obtain a valid IP address. These symptoms may indicate problems with DNS and DHCP services.

- 1. DNS (Domain Name System) :-
- DNS translates domain or hostnames into IP addresses.
- For example, employees can access an application server using its IP address, but DNS allows them to access it using a name such as appserver.company.local.
- If DNS is misconfigured or unavailable, accessing systems by name may fail even when access by IP address works.

- 2. DHCP (Dynamic Host Configuration Protocol) :-
- DHCP automatically assigns IP addresses to devices on the network.
- It also provides network settings such as the subnet mask, default gateway, and DNS server address.
- If DHCP is misconfigured or unavailable, devices may fail to obtain a valid IP address and may be unable to communicate properly with the network.

- 3. Most Likely Misconfiguration :-
- There may be two separate issues:
- DNS issue: Since employees can reach the application server by IP address but not by name, DNS resolution is the most likely cause of that symptom.
- DHCP issue: Since devices sometimes fail to obtain valid IP addresses, DHCP may have a problem with its scope, available leases, configuration, or service availability.
- DHCP can also cause name-resolution problems if it provides an incorrect DNS server address to clients.

- 4. Troubleshooting Steps :-
- Run ipconfig /all to check the device's IP address, DHCP status, and DNS server.
- Run ipconfig /release and ipconfig /renew to request a DHCP lease again.
- Use nslookup appserver.company.local to test DNS resolution.
- Verify that the DHCP scope has available addresses and the correct network settings.
- Check that the DNS service is running and that the application server's DNS record is correct.

SCENARIO :- After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question :- Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

- At Nexus Innovations, employees can access the internet but cannot reach the internal application server after a new firewall is installed. This may happen because the firewall is blocking legitimate internal network traffic.

- 1. Check Network Connectivity
- Verify that the employee computers and application server are powered on and connected to the correct network.
- Check the IP addresses, subnet masks, and default gateways.
- Use the ping command to test connectivity to the server, keeping in mind that a firewall may block ping requests.

- 2. Test the Application Server
- Try accessing the server using its IP address instead of its hostname.
- Test the required application port using PowerShell, for example: Test-NetConnection <server-IP> -Port 443 for HTTPS.
- If access by IP works but the hostname fails, check DNS configuration as well.

- 3. Inspect Firewall Rules and Logs
- Review the firewall's inbound and outbound rules.
- Confirm whether traffic from the employee subnet to the server IP and required application port is being denied.
- Check firewall logs for blocked packets and identify the source IP, destination IP, port, and protocol.
- Verify the correct network zones and routing policies.

- 4. Correct the Firewall Configuration
- Create or modify a rule to allow only the legitimate traffic required to access the application server.
- Specify the correct source network, destination server, protocol, and port.
- Place the rule appropriately according to the firewall's rule-processing order.
- Avoid disabling the entire firewall or allowing all traffic, as this creates security risks.

- 5. Test and Verify
- Test the application from affected employee computers.

- Confirm that authorized users can access the server and that unrelated traffic remains blocked.
- Monitor firewall logs and document the changes.

SECTION B – PRACTICAL APPLICATION

Task 1: Workstation Assembly :- Physically assemble a workstation for the startup office and document the component installation sequence.

Install CPU, RAM, storage, and other core components into the chassis in the correct sequence. Connect all required internal cables (power, data, front panel). Document each installation step in order. System must pass POST (Power-On Self-Test) after assembly.

- Tools and Components Required :-
- Tools: Phillips screwdriver, anti-static wrist strap, and cable ties.
- Components: Computer chassis, motherboard, CPU, CPU cooler, thermal paste (if required), RAM, SSD/HDD, SMPS/PSU, graphics card (if required), and monitor, keyboard, and mouse.
- Procedure: Component Installation Sequence

- Step 1: Prepare the Workspace :-
- I switched off and unplugged the power supply before starting the assembly. I prepared a clean workspace and used an anti-static wrist strap to prevent electrostatic discharge (ESD) from damaging sensitive components.

- Step 2: Install the CPU :-
- I placed the motherboard on an anti-static surface and opened the CPU socket retention mechanism. I aligned the CPU's triangle marker with the corresponding socket marker and carefully placed the processor into the socket without applying force. I then secured the CPU using the retention mechanism.

- Step 3: Apply Thermal Paste and Install the CPU Cooler :-
- I applied a small amount of thermal paste to the CPU if the cooler did not already have pre-applied paste. I installed the CPU cooler according to the manufacturer's instructions and connected its fan cable to the CPU_FAN header on the motherboard.

- Step 4: Install the RAM :-
- I checked the motherboard manual to identify the correct RAM slots. I aligned the notch on the RAM module with the slot and pressed it down until the retaining clips locked into place. For two RAM modules, I used the recommended dual-channel slots.

- Step 5: Install the Storage Drive :-
- I installed the NVMe SSD into the compatible M.2 slot and secured it with the appropriate screw. If using a SATA SSD or HDD, I mounted it in the drive bay and connected the SATA data cable to the motherboard and the SATA power cable from the PSU.

- Step 6: Install the Motherboard into the Chassis :-
- I installed the motherboard standoffs in the correct positions inside the chassis. I fitted the I/O shield if it was not integrated into the motherboard, aligned the motherboard with the standoffs, and secured it using screws without overtightening them.

- Step 7: Install the Power Supply Unit (PSU) :-
- I mounted the PSU in the designated chassis area and secured it with screws. I checked that the PSU was suitable for the system's power requirements.

- Step 8: Install Additional Components :-
- If required, I installed the graphics card into the appropriate PCIe slot and secured it to the chassis. I connected any required PCIe power cables. I also installed additional expansion cards if needed.

- Step 9: Connect Internal Cables :-
- I connected the following cables correctly:
- 24-pin ATX cable: Supplies power to the motherboard.
- 4/8-pin CPU power cable: Supplies power to the CPU.
- SATA power cable: Supplies power to SATA storage drives.
- SATA data cable: Connects SATA storage drives to the motherboard.
- CPU_FAN cable: Powers the CPU cooler fan.
- Front-panel connectors: Connect the power switch, reset switch, and power/HDD LEDs to the correct motherboard headers.
- Front USB and audio cables: Connect the chassis USB ports and audio jacks to the motherboard.
- I routed the cables neatly and ensured that no cables obstructed the cooling fans.

- Step 10: Final Inspection :-

- Before powering on the system, I checked that all components were seated correctly and all required cables were connected. I verified that no loose screws or tools remained inside the chassis.

- Step 11: Connect External Devices :-
- I connected the monitor, keyboard, and mouse to the workstation. I connected the power cable to a suitable power outlet and ensured that the PSU switch was in the OFF position before connecting power.

- Step 12: Power On and Perform POST :-
- I switched on the PSU and pressed the workstation's power button. I checked whether the fans were spinning, the motherboard displayed any diagnostic indicators, and the monitor showed the manufacturer's logo or BIOS/UEFI screen.
- I entered the BIOS/UEFI setup to verify that the CPU, installed RAM, and storage drive were detected correctly. I checked for error messages or abnormal hardware indicators.

- As a result :-
- The workstation was assembled successfully, all required internal cables were connected, and the system passed POST. The CPU, RAM, and storage were detected in the BIOS/UEFI setup, confirming that the essential hardware components were installed correctly.

- Precautions :-
- Always disconnect power before installing or removing components.
- Use ESD protection when handling the motherboard, CPU, and RAM.
- Never force the CPU or RAM into their sockets.
- Apply thermal paste only when required by the cooler's instructions.
- Ensure the CPU power, motherboard power, and front-panel cables are connected correctly.
- Follow the motherboard and component manufacturers' installation instructions.
- Keep cables away from fans and verify all connections before powering on.

Task 2: RAM Mismatch :- Troubleshooting ASSESSMENT FILE CompTia N+ M2-A1 Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings on a startup office workstation.

Identify the RAM mismatch error via BIOS/UEFI POST messages. Determine the root cause (mismatched speed, timing, or capacity).

Resolve the mismatch by reconfiguring or reseating the RAM correctly. Confirm the system boots normally after the fix.

- Tools Required :-
- Startup office workstation
- Compatible RAM modules
- Phillips screwdriver
- Anti-static wrist strap
- Motherboard manual

- Step 1: Identify the RAM Mismatch Error :-
- I powered on the workstation and observed the POST messages displayed on the monitor. The system indicated a memory configuration error, or the BIOS/UEFI showed that the installed RAM was not operating as expected.
- I entered the BIOS/UEFI setup by pressing the appropriate key, such as Delete, F2, or another key specified by the manufacturer.

- Step 2: Check the RAM Configuration :-
- Inside the BIOS/UEFI settings, I checked the following details:
- Total installed RAM capacity.
- RAM speed and supported frequency.
- Memory configuration and detected modules.
- Any memory-related warning or error messages.
- I compared these details with the motherboard specifications and the RAM manufacturer's specifications.

- Step 3: Determine the Root Cause :-
- I checked whether the mismatch was caused by one of the following factors:
- Mismatched RAM speed: The installed RAM modules had different rated speeds, or the selected memory profile was unsupported.
- Mismatched timing: The configured memory timings were not supported by the installed modules or the memory controller.
- Mismatched capacity: The RAM modules had different capacities, or the total installed RAM exceeded the motherboard's supported capacity.
- Improper installation: A RAM module was not fully seated in its slot.

- Step 4: Resolve the RAM Mismatch :-
- I switched off the workstation, disconnected the power cable, and used anti-static precautions before handling the RAM modules.
- I then performed the appropriate corrective action:
- Reseated the RAM modules securely in the recommended motherboard slots.
- Checked the motherboard manual to confirm RAM compatibility and the recommended slot arrangement.

- If necessary, used compatible RAM modules with supported capacities and speeds.
- Entered BIOS/UEFI and restored the memory settings to Auto or default settings.
- Disabled an unsupported XMP/EXPO memory profile if it caused instability.
- Saved the settings and restarted the workstation.
- I avoided manually increasing RAM voltage or changing timings without confirming the manufacturer's specifications.

- Step 5: Verify the Fix :-
- After correcting the RAM configuration, I restarted the workstation and observed the POST process.
- I entered BIOS/UEFI again and confirmed that the expected RAM capacity was detected and that the memory operated at a supported speed. I then allowed the computer to boot into the operating system and checked that it worked normally without memory-related error messages.
- If necessary, I would run a memory diagnostic test to confirm stability.

- Result :-
- The RAM mismatch issue was addressed by checking compatibility, correcting the RAM installation or configuration, and verifying the system's memory detection. After the fix, the workstation completed POST and booted normally without the previous memory error.

- Precautions :-
- Always switch off and unplug the workstation before reseating RAM.
- Use anti-static protection when handling memory modules.
- Check motherboard and CPU memory compatibility before installing RAM.
- Never force a RAM module into a slot.
- Use supported memory speeds, timings, and voltage settings.
- Confirm the final configuration in BIOS/UEFI after restarting.

Task 3: LAN Topology :- Build Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

Use Cisco Packet Tracer to build the topology. Include 5 end-systems and 1 network printer. Assign IP addresses from a Class C subnet to all devices. All assigned IPs must be reachable via Ping.

- Tools Required :-
- Cisco Packet Tracer
- 1 Cisco 2960 switch
- 5 PCs
- 1 network printer

- Copper straight-through Ethernet cables

- Step 1: Build the Network Topology :-
- Open Cisco Packet Tracer.
- Add one Cisco 2960 switch to the workspace.
- Add five PCs and one printer.
- Connect each PC and the printer to the switch using copper straight-through cables.
- Wait until the switch ports show active links.

- Step 2: Assign Class C IP Addresses :-
- I selected the private Class C network 192.168.1.0/24.
- Subnet mask: 255.255.255.0

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.10	255.255.255.0

- All devices use unique IP addresses in the same subnet, so a default gateway is not required for communication within this LAN.

- Step 3: Configure IP Addresses on PCs :-
- For each PC:
- Click the PC.
- Open the Desktop tab.
- Select IP Configuration.
- Select Static.
- Enter the assigned IP address and subnet mask.
- Repeat the process for all five PCs.

- Step 4: Configure the Printer :-
- Click the printer in Packet Tracer.
- Open its Config or network configuration section, depending on the device model.
- Select the FastEthernet interface or IP configuration option.
- Assign IP address 192.168.1.10.
- Set the subnet mask to 255.255.255.0.

- Step 5: Verify Connectivity Using Ping :-
- Click PC0.
- Open Desktop → Command Prompt.
- Enter the following command:
- ping 192.168.1.2
- Test connectivity to the remaining PCs and printer using their IP addresses.
- Example commands:
- ping 192.168.1.2
- ping 192.168.1.3
- ping 192.168.1.4
- ping 192.168.1.5
- ping 192.168.1.10
- Repeat the tests from other PCs if required. Successful replies indicate that the devices can communicate with one another. If the first ping times out, repeat it because initial ARP resolution may cause a delay.

- Step 6: Troubleshooting :-
- If ping fails, I would:
- Check the physical cable connections and switch port status.
- Verify that all IP addresses are unique and correctly entered.
- Confirm that all devices use the same subnet mask.
- Check the printer's network interface configuration.
- Ensure that the devices are connected to the same LAN and that no access-control or security setting blocks communication.

- Result :-
- A LAN topology containing five PCs and one network printer was created in Cisco Packet Tracer. Class C private IP addresses were assigned to all six devices. Connectivity was tested using Ping commands, and successful replies confirmed that the devices were reachable within the LAN.

- Precautions :-
- Assign a unique IP address to every device.
- Use the correct subnet mask for all devices.
- Connect each device to the switch using the correct Ethernet interface.
- Verify connectivity using Ping before declaring the practical successful.

Task 4: Clean OS Installation :- Perform a clean OS installation with manual disk partitioning and startup optimization on a startup office workstation.

Perform a clean install of Windows/Linux from ISO. Manually partition the disk (not guided/automatic partitioning). Optimize startup programs/services for faster boot. OS must boot without errors after installation.

- Tools and Requirements :-
- Workstation or compatible virtual machine
- Windows or Linux ISO image
- Bootable USB drive
- USB creation tool, such as Rufus
- Backup of important data
- Internet connection for updates and drivers, if required

- Step 1: Prepare the Installation Media :-
- Download the required Windows or Linux ISO image from the official website.
- Create a bootable USB drive using a suitable tool.
- Back up all important files because a clean installation may erase existing data.
- Confirm that the workstation meets the operating system's hardware requirements.

- Step 2: Boot from the USB Drive :-
- Insert the bootable USB drive into the workstation.
- Restart the computer and open the boot menu using the manufacturer's designated key.
- Select the USB drive in the appropriate UEFI or legacy boot mode.
- Start the operating system installation process.

- Step 3: Start the Clean Installation :-
- Select the required language, time, and keyboard settings.
- Start the installation process.
- Select the custom installation option, such as Custom: Install Windows only (advanced) in Windows Setup.
- Select the target disk carefully and proceed to manual partitioning.

- Step 4: Manually Partition the Disk :-
- I inspected the target disk and created partitions according to the selected operating system and firmware mode.
- Example partition layout for a 256 GB Windows workstation:

- Partition	- Approximate Size	- Purpose
- EFI System Partition	- 260 MB	- Stores UEFI boot files

Partition	Approximate Size	Purpose
Microsoft Reserved (MSR)	16 MB	Reserved for Windows disk management
Windows (C:)	200 GB	Operating system and applications
Recovery partition	1 GB	Windows recovery environment
Data (D:)	Remaining space	User files and work documents

- The partition sizes are examples and can vary depending on Windows requirements and the disk capacity. The EFI, MSR, and recovery partitions should be created according to the Windows installer and system requirements. The data partition uses NTFS.
- For Linux, a typical manual layout may include an EFI System Partition for UEFI systems, a root (/) partition, and an optional separate /home partition or swap space, depending on the distribution and requirements.
- Important: I confirmed the target disk before deleting or formatting any partition, because these operations can permanently erase data.

- Step 5: Install the Operating System :-
- Select the intended OS partition as the installation destination.
- Format the target OS partition if required and confirm the installation.
- Allow the installer to copy files and install system components.
- Restart the workstation when prompted and remove the USB drive when appropriate.
- Complete the initial setup, including the user account, region, and network settings.

- Step 6: Install Drivers and Updates :-
- Check whether essential hardware devices are detected correctly.
- Install required chipset, network, graphics, and other drivers from trusted sources.
- Install available operating system updates.
- Restart the workstation if required.

- Step 7: Optimize Startup Programs and Services :-
- For Windows:
- Open Task Manager → Startup apps.
- Review applications that launch automatically.
- Disable unnecessary non-essential startup applications.
- Open Settings → Apps → Startup if needed to review startup entries.
- Avoid disabling essential security, hardware, or system services.

- Step 8: Verify the Installation :-
- Restart the workstation.
- Confirm that the operating system boots to the login screen or desktop without errors.
- Verify that the CPU, RAM, storage, keyboard, mouse, and network adapter are detected.
- Check Device Manager in Windows or appropriate system tools in Linux for hardware or driver errors.
- Confirm that the system partitions are present and that the startup applications have been optimized.
- Restart once more to ensure that the system remains stable.

- Result :-
- A clean operating system installation was performed using bootable installation media. The disk was manually partitioned, necessary drivers and updates were installed, and unnecessary startup applications were disabled. The workstation was restarted and checked to verify that the operating system booted without errors.

- Precautions :-
- Back up important data before installation.
- Verify the correct target disk before formatting or deleting partitions.
- Use an official ISO image and trusted installation tools.
- Use the correct partition scheme and boot mode, such as GPT with UEFI for a typical modern Windows installation.
- Do not disable essential operating system, security, or hardware services during optimization.
- Verify successful booting and hardware operation before completing the assessment.

SECTION C – MINI CAPSTONE PROJECT

Mini Project: Nexus Innovations — Diagnosing Random System Reboots Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads.

The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a

remediation plan, combining hardware, power, and troubleshooting concepts from this module.

Your project must: Identify the three hardware components most likely to cause random reboots under load Explain why the systems pass POST in the morning but fail in the afternoon. Describe how you would use a stress-test tool to confirm a thermal or power issue. Propose three specific steps to remediate the frequent system restarts. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

- 1. Introduction :-
- Nexus Innovations has three rendering workstations that pass POST (Power-On Self-Test) every morning but restart randomly during heavy afternoon workloads. The computers are installed inside poorly ventilated desk cabinets, which may cause overheating and restrict airflow.
- The most likely causes include excessive heat, insufficient power delivery, and unstable hardware under heavy workloads. A systematic troubleshooting process is required to identify the root cause and prevent further system restarts.

- 2. Three Hardware Components Most Likely to Cause Random Reboots :-
- A. CPU (Central Processing Unit)
- The CPU performs calculations and processes instructions during rendering tasks. Under heavy workloads, its temperature and power consumption increase significantly. Poor ventilation, a faulty CPU cooler, accumulated dust, or improperly applied thermal paste may cause the CPU to overheat.
- When the CPU reaches its thermal limits, the system may reduce performance or shut down to protect the hardware. This can contribute to unexpected restarts.
- B. PSU (Power Supply Unit)
- The PSU supplies electrical power to the motherboard, CPU, graphics card, storage drives, and other components. During rendering, the CPU and GPU may consume much more power than they do during light workloads.
- If the PSU has insufficient capacity, is defective, or cannot provide stable power under load, the workstation may restart unexpectedly. Loose power connectors can also contribute to instability.
- C. GPU (Graphics Processing Unit)
- Rendering software places a heavy load on the GPU, increasing its temperature and power consumption. If the GPU cooling system is blocked by dust or has a faulty fan, it may overheat.
- A GPU that draws more power than the PSU can reliably supply may also cause system instability or sudden restarts.

- RAM instability, motherboard faults, and other hardware problems should also be considered if the primary checks do not identify the cause.

- 3. Why the Systems Pass POST in the Morning but Fail in the Afternoon :-
- POST checks essential hardware components during startup. In the morning, the workstations are relatively cool, and the system may only be performing light tasks. Therefore, they can pass POST successfully.
- During afternoon rendering workloads, the CPU and GPU generate significantly more heat and consume more power. Poor ventilation inside the desk cabinets prevents hot air from escaping and reduces the effectiveness of the cooling system.
- As temperatures rise, the system may experience thermal throttling, instability, or protective shutdowns. At the same time, increased power demand may expose a weak or failing PSU.
- Therefore, passing POST does not guarantee that a workstation will remain stable during sustained, high-load operation.

- 4. Using a Stress-Test Tool to Confirm Thermal or Power Issues :-
- I would use a trusted hardware monitoring and stress-testing tool, such as OCCT, to investigate the problem.
- Procedure
- Inspect the workstation for dust, blocked air vents, loose power connectors, and faulty cooling fans.
- Record the CPU and GPU temperatures, fan speeds, and system behaviour while the computer is idle.
- Run a controlled CPU stress test while monitoring temperatures and system stability.
- Run a GPU stress test to check its temperature and power behaviour during rendering-like workloads.
- If necessary, run a combined CPU and GPU test to increase overall system power demand. This should be performed under supervision while monitoring temperatures and stopping immediately if unsafe conditions occur.
- Review the monitoring results and system logs for thermal warnings, hardware errors, or unexpected shutdowns.
- Interpretation of Results
- Thermal issue: If CPU or GPU temperatures rise excessively and the system becomes unstable, overheating or inadequate cooling may be responsible.
- Possible PSU issue: If the system restarts under combined CPU and GPU load without clear evidence of overheating, power delivery should be investigated.
- Other hardware issue: If temperatures and power delivery appear normal, check RAM stability, motherboard components, drivers, and system logs.
- A stress test alone cannot prove that a PSU is faulty. Further checks, such as verifying PSU specifications, inspecting connectors, and testing with a known-good suitable PSU, may be required.

- 5. Three Specific Remediation Steps :-
- Step 1: Improve Cabinet Ventilation and Cooling :-
- Move the workstations out of enclosed desk cabinets or improve airflow by adding suitable ventilation. Clean dust from heatsinks and fans, verify CPU and GPU fan operation, and ensure the CPU cooler is correctly installed with appropriate thermal paste.
- Justification: Better airflow and effective cooling reduce component temperatures and help prevent thermal instability during sustained rendering workloads.
- Step 2: Inspect and Verify the Power Supply :-
- Check the PSU's rated wattage, quality, age, and suitability for the installed CPU and GPU. Inspect the 24-pin motherboard connector, CPU power connector, and GPU power connectors. Replace a defective or inadequate PSU with a suitable model if testing confirms a power-delivery problem.
- Justification: A suitable PSU provides reliable power during periods of high demand and reduces the risk of instability caused by inadequate or faulty power delivery.
- Step 3: Perform Controlled Testing and Monitor Stability :-
- Update relevant BIOS/UEFI firmware and hardware drivers where appropriate, check RAM stability, and run controlled CPU and GPU stress tests after correcting cooling and power issues. Monitor temperatures, fan speeds, and system logs. Verify that each workstation remains stable under a representative rendering workload.
- Justification: Testing after remediation confirms whether the corrective actions resolved the issue and helps identify any remaining hardware or software faults.

- 6. Technical Justification: Is a PSU Upgrade Necessary? :-
- A PSU upgrade may be necessary if the existing unit has insufficient wattage, poor quality, deteriorated components, or cannot reliably support the combined CPU and GPU power demand.
- However, a PSU upgrade should not be performed automatically. If the main cause is overheating due to poor ventilation, replacing the PSU alone will not solve the problem. The existing PSU may be adequate if it is functioning correctly and has sufficient capacity for the installed components.
- The IT team should inspect the PSU specifications, evaluate the system's power requirements, and test power delivery before deciding whether an upgrade is required.

- 7. Conclusion :-
- The most likely causes of random restarts at Nexus Innovations are CPU overheating, GPU overheating, and inadequate or unstable PSU power delivery. The workstations pass POST in the morning because they are cooler and under less load, but heavy afternoon rendering increases heat generation and power demand.
- By improving cabinet ventilation, inspecting cooling systems, verifying PSU suitability, and conducting controlled stress tests, the IT team can identify the root cause and implement an effective solution. These measures should improve workstation stability, protect hardware, and reduce interruptions to rendering work.

SECTION D – AI – AUGMENTED LEARNING

STEP 1 · BUILD WITH AI :- Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer. The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address. The plan must allocate addresses for all 5 systems and 1 printer named in the scenario. Instruct the AI to also identify which addresses to reserve for future expansion. Document the exact prompt(s) used.

STEP 2 · TEST & DEBUG (WITHOUT AI) :- Then, working without AI, verify the plan for errors — overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself. The exact prompt(s) you gave the AI tool. The AI's original subnetting plan and your corrected version. All items required 3–4 line note explaining what you changed and why the AI's version needed it.

- STEP 1: BUILD WITH AI :-
- 1. Objective :-
- The objective of this task is to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network using an AI tool. The network must support five computer systems and one network printer, with reserved addresses for future expansion.

- 2. Exact Prompt Used :-
- AI Tool Used: ChatGPT
- Prompt:
- "Generate an IP addressing and subnetting plan for Nexus Innovations' startup office network containing five computer systems and one network printer. Specify the network subnet, subnet mask, usable host range, broadcast address, and an individual IP address for each device. Reserve some IP addresses for future expansion. Present the plan in a table and ensure that all devices use the same subnet."

- 3. AI-Generated Original IP Addressing Plan :-

- The AI generated the following sample plan using the private Class C network 192.168.1.0/29.
- Subnet mask: 255.255.255.248

Parameter	AI-Generated Value
Network address	192.168.1.0
Subnet mask	255.255.255.248
Usable host range	192.168.1.1 – 192.168.1.6
Broadcast address	192.168.1.7

- Device allocation:

Device	Assigned IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5
Network Printer	192.168.1.6

- Future expansion: No usable host addresses remain available in this subnet after assigning the five PCs and the printer.

- STEP 2: TEST AND DEBUG WITHOUT AI :-
- 4. Manual Verification of the AI Plan :-
- I manually checked the subnet mask, network address, usable host range, broadcast address, device allocations, and available host capacity.
- For a /29 subnet:
- Total addresses = $2^{32-29} = 8$
- Network address = 1
- Broadcast address = 1
- Usable host addresses = $8 - 2 = 6$
- The original plan correctly identified the network address, subnet mask, broadcast address, and usable host range. However, all six usable addresses were assigned to the five computers and one printer, leaving no space for future expansion.

- 5. Corrected IP Addressing Plan :-
- To allow future growth, I selected the 192.168.1.0/28 subnet.
- Subnet mask: 255.255.255.240

- Parameter - Corrected Value
- Network address - 192.168.1.0
- Subnet mask - 255.255.255.240
- Usable host range - 192.168.1.1 – 192.168.1.14
- Broadcast address - 192.168.1.15
- Total addresses - 16
- Usable host addresses - 14
- Corrected device allocation:-
- Device - Assigned IP Address
- PC0 - 192.168.1.1
- PC1 - 192.168.1.2
- PC2 - 192.168.1.3
- PC3 - 192.168.1.4
- PC4 - 192.168.1.5
- Network Printer - 192.168.1.6
- Reserved for future expansion:-
- 192.168.1.7 – 192.168.1.14: Eight available host addresses reserved for future computers, printers, or other network devices.

- 6. Manual Subnetting Calculation :-
- For the corrected /28 subnet:
- Host bits = 32 − 28 = 4
- Total IP addresses = $2^4 = 16$
- Usable host addresses = 16 − 2 = 14
- Subnet mask = 255.255.255.240
- Network address = 192.168.1.0
- Broadcast address = 192.168.1.15
- Usable host range = 192.168.1.1 to 192.168.1.14
- The corrected subnet supports all six required devices and leaves eight additional usable IP addresses for future expansion.

- 7. Note: What I Changed and Why :-

- The original AI-generated plan used a /29 subnet, which provides only six usable host addresses. Although the six devices could be assigned valid IP addresses, no addresses remained for future expansion. I corrected the plan to use a /28 subnet, which provides fourteen usable host addresses. This change preserves the existing device allocations and reserves eight addresses for future network growth.

- 8. Final Result :-
- The original AI-generated plan was checked manually for subnetting errors and host capacity. The corrected plan uses the 192.168.1.0/28 subnet with the subnet mask 255.255.255.240, a usable range of 192.168.1.1–192.168.1.14, and broadcast address 192.168.1.15. All five computers and the network printer have unique IP addresses, and eight usable addresses remain reserved for future expansion.
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